

Airfoil Decambering for Gust Load Alleviation Using a Bi-stable Hinge

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Wing sections with bi-stable hinges outfitted on trailing-edge flaps provide a promising means for passively rejecting gust loads on an aircraft. This study found that there exists a two-way coupling between the aerodynamic loads experienced by the airfoil and the deflection of the bi-stable hinge and corresponding flap. The current work successfully demonstrates how the energy wells of a bi-stable hinge system can be configured to introduce decambering of the airfoil when stall limits are approached. Additional contributions of the final paper will include further development of methods aimed to predict the two-way coupling between aerodynamic loading and bi-stable hinge deflection, as well as further assessment of gust load alleviation strategies.

I. Nomenclature

α	=	Angle of attack
c	=	Chord length
c_f	=	Flap chord length
C_d	=	Sectional drag coefficient
C_h	=	Sectional hinge moment coefficient
C_l	=	Sectional lift coefficient
δ	=	Flap deflection angle
GLA	=	Gust Load Alleviation
H'	=	Hinge moment per unit span
M	=	Mach number
MAV	=	Micro Air Vehicles
q_∞	=	Freestream dynamic pressure
Re	=	Reynolds number
x	=	Length along streamwise coordinate
z	=	Length along stream-normal coordinate

II. Introduction

GLOBAL CO₂ emissions by the aviation sector were estimated to be 920 million tons in 2019, an increase of approximately 2-3% from 2018 [1, 2]. One contribution to aircraft CO₂ emissions is in-flight gusts and turbulence as aircraft are rerouted to avoid turbulent regions, thus leading to increased flight path distances and increasing fuel burned [3]. As the climate continues to change, the frequency of wind shear and clear-air turbulence at cruising altitudes is increasing [4]. Subsequently, flight paths will increase to avoid the turbulent regions, increasing the fuel burned and further increasing CO₂ emissions.

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In-flight gust load alleviation (GLA) systems can increase aircraft efficiency and reduce the environmental impact of aircraft operation. GLA systems respond to sudden dynamic loads and reduce the bending stresses experienced by the airframe [5]. In addition to reducing the re-routing distance, GLA systems allow for a reduction in structural weight, which directly translates to reduced fuel consumption and lower emissions [3, 6]. Active gust load alleviation systems have been successfully implemented on aircraft such as the Lockheed C-5A, Airbus A320, and Boeing 787 [7, 8]. These aircraft implement control systems that detect gusts and output the corrections to the flight computer to deflect the control surfaces. For the C-5A, the GLA system uses symmetric aileron deflections to redistribute the spanload resulting in a wing load reduction of 32-52% [7, 9]. However, the sensors required to detect gusts can add additional weight to the aircraft, which reduces their benefits. More significantly, these systems are limited by sensor and actuator bandwidth [5].

Much of the recent work on gust load alleviation has focused on Micro Air Vehicles (MAV) because their small size, low mass, and operation in turbulent urban environments make them particularly vulnerable to transverse gusts. Transverse gusts are gusts perpendicular to the freestream velocity and pose a danger to MAVs by causing unanticipated extreme structural loading and possible loss of control [10]. A recent study by Andreu-Angulo et al. investigated two different ways of simulating gust in experiments: a top-hat discontinuous and a sine-squared velocity profile [10]. They found that the gust shape is a determining factor in gust encounter characteristics. Li et al. investigated the effects of deploying a trailing edge flap in a top-hat type gust for various gust ratios in order to validate their low-order model [11]. Gust ratio (GR) is the ratio of the gust velocity to the freestream velocity: $GR = V_g/U_\infty$. The study found that deploying the flap successfully mitigated the C_l for low gust ratios of 0.5 where the flow remained mostly attached, while only marginally mitigated the C_l for a gust ratio of 0.75 as the flow was highly separated.

In addition to control surface deflections, other mechanisms for GLA have been investigated. Al-Battal et al. sought to reduce extreme loads using blown jets [12]. Surface normal and upstream jet blowing near the leading and trailing edge were investigated. It was found that trailing edge actuation locations were more effective for low angles of attack and leading edge actuation locations were more effective at high angles of attack. For the leading edge blowing case, a maximum reduction of $\Delta C_l = -0.33$ was achieved at the stall angle of attack.

Active GLA systems add weight and complexity to the aircraft through the addition of necessary sensors, actuators, and flight controllers. One solution to this may be the use of GLA systems implemented through passive airfoil camber morphing using multi-stable materials. Multi-stable materials have two or more stable states and are made by defining how the material reacts to different energy inputs [13]. Diaconu et al. studied how a bi-stable plate could be used for changing the camber and chord of an airfoil section using carbon-fiber laminated composites to achieve bistability through thermal curing [14]. These materials would snap-through from one stable state to the next at set temperatures, however additional internal structure would be required to support in-flight loads, thus reducing the benefit of the morphing airfoil. Additionally, because the composite was thermally cured, the performance of the structure was dependent on environmental factors such as temperature and humidity. They found that at for high temperatures and high moisture absorption the bistability of the plate was largely diminished. Bilgen et al. considered the use of piezoelectric actuators as the forcing mechanism for a morphing bi-stable wing. They found that the composite wing was stiff enough to support aerodynamic loads and that dynamic snap-through could be achieved in adverse pressure gradients [15]. Namdeo et al. tested a bi-stable trailing edge that would snap-through due to perceived pressure fields [16]. The goal of this study was to capture the dynamic loading during the snap-through process using motion tracking cameras to characterize snap-through.

The current work seeks to resolve the coupling of a bi-stable hinge and experienced aerodynamic loads in order to produce a passive GLA system. For a selected loading hinge moment, the flap will deflect to passively decamber the airfoil, where the flap hinge moment serves as the snap-through forcing mechanism. Since the bi-stable hinge experiences an initial deflection prior to snap-through due to the non-zero hinge moments experienced, which in turn influences the deflection angle, this project aims to resolve the two way coupling of the bi-stable hinge and the aerodynamic forces.

III. Design Methodology

A. Design Point Selection

An NLF-0414F airfoil, shown in Fig. 1a, was selected due to the significant role served by the trailing-edge flap to regulate the cruise lift conditions across a high-efficiency drag bucket. As shown in Fig. 1b, the hinge moment decreases linearly with angle of attack. This was desirable for the bi-stable hinge configuration, as each value of hinge

moment coefficient corresponds to a unique angle of attack for a given flap deflection angle. This characteristic of the hinge moment allowed for a better prediction of at which angle of attack the snap-through would occur. For this study, the hinge moment coefficient is defined as

$$C_h = \frac{H'}{q_\infty b c_f^2} \quad (1)$$

where c_f is the chord length of the flap element.

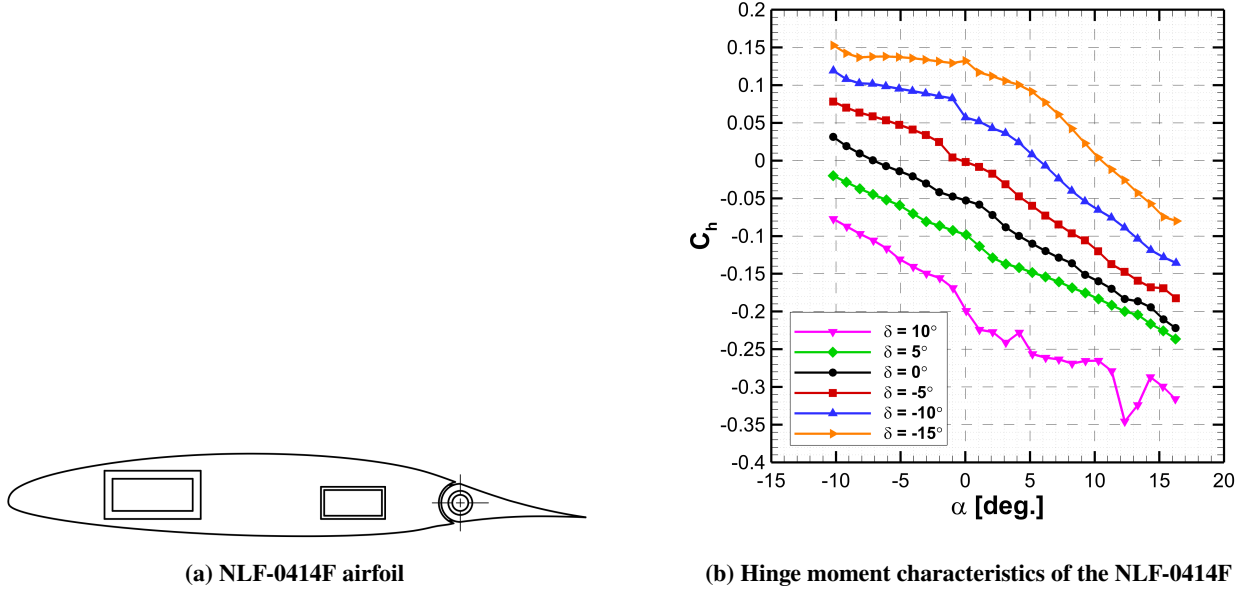


Fig. 1 Selected NLF-0414F airfoil hinge moment characteristics

The snap-through hinge moment was selected such that the flap would deflect $\delta = -5^\circ$ and would maintain the C_l experienced at the previous angle of attack, prior to the snap-through. Such an approach resulted in a bi-stable hinge that could passively trim the airfoil to maintain a fixed C_l value when experiencing an increase in angle of attack. A dimensional hinge moment of 2.2 lb-ft ($C_h = -0.15$) corresponding to a C_l of 1.14 was selected as the target for snap-through, representing the pre-stall condition where the decambering would alleviate some of the trailing edge separation of the baseline $\delta = 0^\circ$ case.

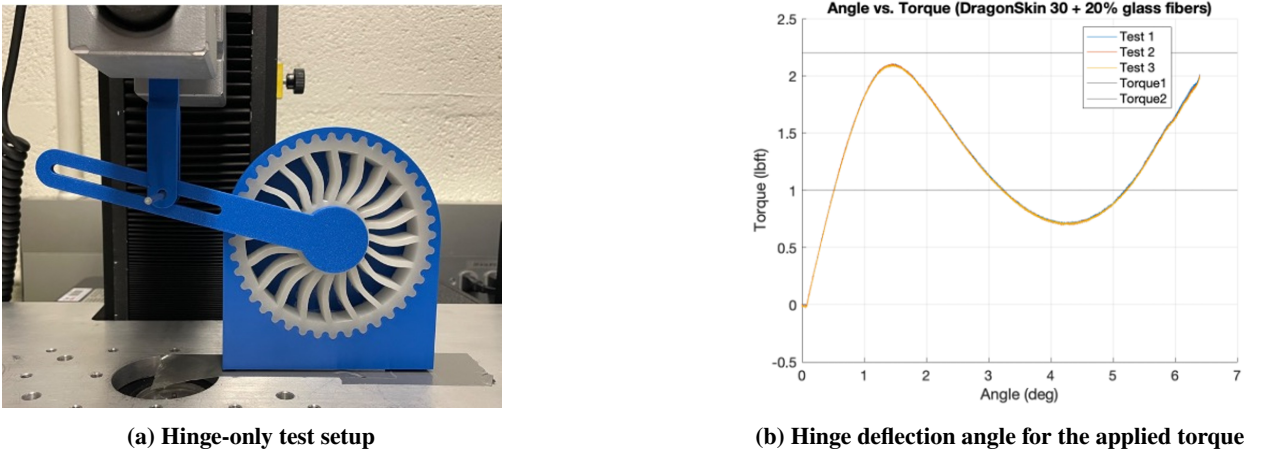


Fig. 2 Hinge-only test setup and results

B. Bi-stable Hinge Design

The bi-stable hinge was constructed with a silicon rubber DragonSkin 30 material with 20% glass fibers. A 5:1 gearbox was required to reduce the hinge deflection to the desired flap deflection angle. The outer diameter of the hinge was fixed in place while the inner diameter was connected to the flap spar to facilitate the flap deflection. Prior to integration with the wind tunnel model, hinge-only tests were conducted. Figure 2 shows the setup of the hinge-only prototype and the output torques for each deflection angle. In Fig. 2b, the region of the hinge deflection profile corresponding to 1.5° to 4.5° is unstable and would not be expected to occur for actual use of the hinge. Because of the bi-stability of the hinge, once the 2.2 [lb-ft] torque is reached the hinge snaps to the next stable position.

C. Experimental Setup

A 2D model of a NLF-0414F airfoil, originally designed by Jackson, was utilized for this experiment and can be seen in Fig. 3b [17]. The model was composed of two elements; a main element and a flap element. The main element had two rectangular steel spars located at $x/c = 0.25$ and $x/c = 0.60$ and the flap element had a circular spar located at the flap hinge line of $x/c = 0.779$ and $z/c = 0.0049$. The leading edge of the flap was located at $x/c = 0.75$. The model had a chord length of 1.5 ft and a span of 2.8 ft and was constructed of a carbon fiber skin surrounding a foam core.



(a) CAD model with bi-stable hinge



(b) Installed model in wind tunnel

Fig. 3 CAD and installation of NLF-0414F airfoil model

Surface pressure distribution data were collected from 92 surface pressure taps. The model consisted of 60 surface pressure taps on the main element, 21 taps on the flap element, and 11 spanwise taps on the main element at $x/c = 0.65$. The taps on the main element were angled at a spanwise wedge angle of 15° in order to prevent possible influences due to turbulent flow generated by the taps upstream.

IV. Experimental Procedures

The tests were conducted in the open-return subsonic wind tunnel at the Aerodynamics Research Laboratory, University of Illinois Urbana-Champaign. All measurements were acquired for a fixed Reynolds number of $Re = 1.8 \times 10^6$ ($M_\infty = 0.17$), which was controlled within a 0.5% margin. The model was mounted vertically wall-to-wall in the 2.8 ft $\times$ 4 ft rectangular test section, as shown in Fig. 3b. The model was mounted to the ATE Aerotech 3-component external force balance which collected normal, axial, and moment data which were used to calculate the lift, drag, and pitching moment coefficients. The force balance also allowed for angle of attack control. Three ESP-32HD total pressure transducer modules were used to collect the model surface pressure distribution. Each pressure tap on the

model was connected to a port on a pressure transducer via a vinyl tube. The pressure transducers converted the pressure differential between the local pressure of each port and the static freestream pressure into a voltage, which was sent to the Digital Temperature Compensation (DTC) Initium data acquisition system. The DTC Initium then transmitted the voltage signals to the data acquisition computer. Hinge moment data were collected from an inline Omega LCF-25 load cell connecting the flap spar and the flap deflection mechanism. Prior to testing, the load cell was calibrated by applying known hinge moments by hanging weights from the flap and recording the output voltage.

A wake rake system was used to acquire the wake defect produced downstream of the airfoil model. The wake rake consisted of 59 total pressure probes aligned parallel with the freestream direction. For angles of attack in the linear lift range, the span of the wake rake was sufficient to capture the entire wake profile, whereas when the flow was separated, the wake rake was traversed in the chord-normal direction to ensure the entire wake profile was captured. Drag was then calculated following the methods described by Ansell [18].

A. Baseline Rigid Deflections

Baseline data were acquired for the airfoil flap at rigid deflection angles of $\delta = 10^\circ$ and $\delta = -15^\circ$ to 5° in increments of 1° . For this paper, "rigid deflection" refers to cases where the flap deflection angle was set prior to each run and held constant during the run. Figure 4 shows the flap actuation method. The flap deflections were set manually before each run using a Zaber T-LSR450B linear traverse connected to the flap spar. The inline load cell used to measure hinge moments was attached to the linkages connecting the flap spar to the linear traverse. The polar was then taken for angles of attack of $\alpha = -10^\circ$ to 16° in order to capture stall.

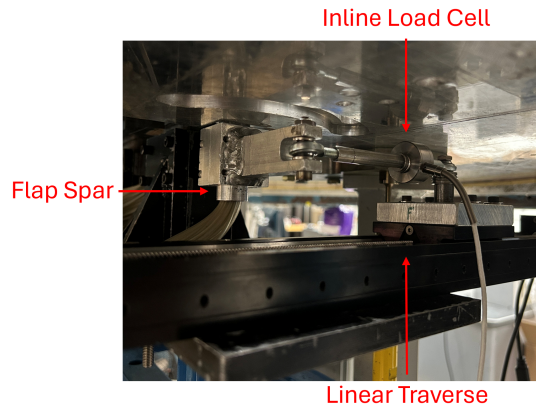


Fig. 4 Baseline rigid flap deflection experimental setup

B. Bi-stable Hinge Deflections

The bi-stable hinge was attached to the flap spar using the same linkage system as the rigid cases with the inline load cell and can be seen in Figs. 3a, 4, and 5. Additionally, the flap deflection angle was measured using a Dynapar HS35R optical rotary encoder. Measurements were also acquired with the bistable hinge installed across a range of angles of attack of $\alpha = -10^\circ$ to 16° .

C. Validation

The experimental setup was validated by comparing with past results from Lee [19] at $Re = 1.8 \times 10^6$ and $\delta = 0^\circ$ and are shown in Fig. 10 in the Appendix. While the moment coefficient data did not show strong agreement, since the balance and pressure moments aligned better for this current project, it was determined that the difference was allowable.

V. Results

The aerodynamic performance of the airfoil with the bi-stable hinge installed can be seen in Figs. 6 and 7. Here, forward loading refers to increasing the applied hinge moment by increasing α until snap-through occurs. The two-way coupling of the bi-stable hinge and aerodynamic forces manifests as a "flattening" of the aerodynamic forces through a

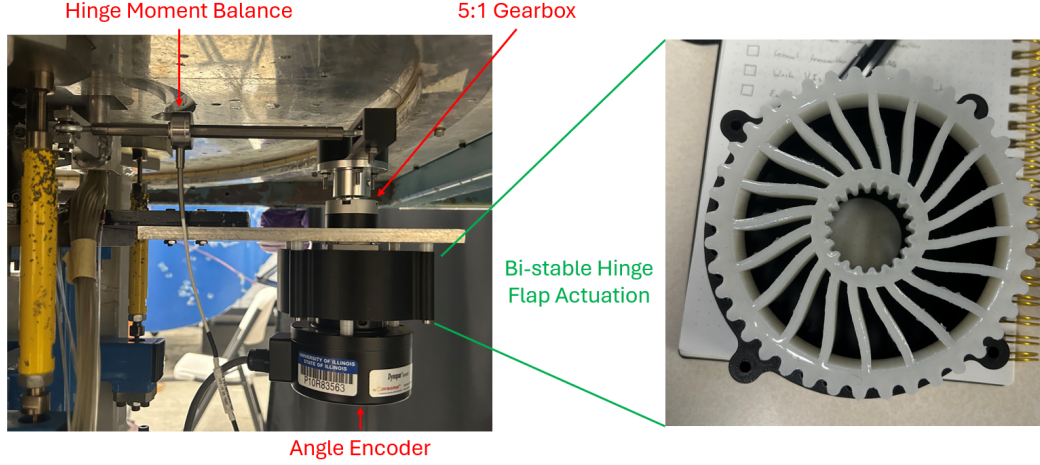
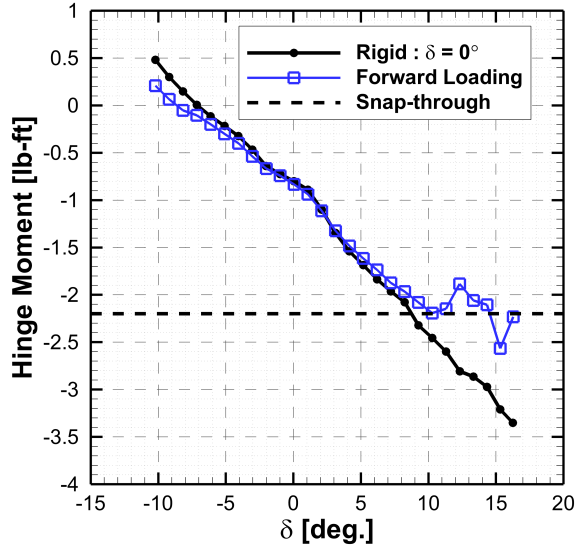
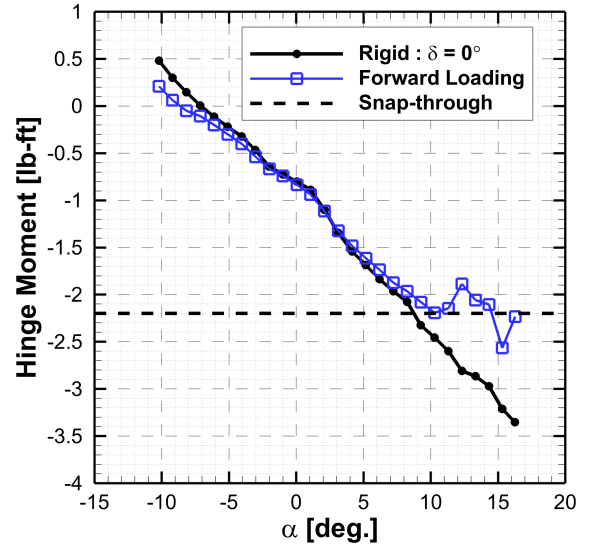


Fig. 5 Bi-stable hinge experimental setup

reduction of the slope. This is due to the initial deflection of the hinge prior to snap-through. This initial deflection in turn decreases the hinge moment, which decreases the deflection, which influences the hinge moment, and so on. As shown in Fig. 6, the flap has a positive deflection for negative α and a negative deflection for positive α . This is because the hinge was installed such that $\delta = 0^\circ$ occurred at $\alpha = 0^\circ$. Therefore, the bi-stable results align best near $\alpha = 0^\circ$. Prior to the snap-through of the hinge, angles of attack farther from $\alpha = 0^\circ$ result in greater deflections from the rigid-flap $\delta = 0^\circ$ deg results. Figure 6b shows that a maximum initial deflection of $\delta = -1.01^\circ$ occurs just prior to snap-through, which occurs at an applied torque of $H' = -2.2$ lb-ft. The exact point where this hinge moment was reached is not captured in the data as snap-through occurred while the model was changing angles of attack from $\alpha = 10^\circ$ to $\alpha = 11^\circ$.



(a) Hinge moment of bi-stable hinge test compared to rigid $\delta = 0^\circ$ deflection



(b) Dimensional hinge moment deflection comparison with hinge-only tests

Fig. 6 Coupling of the bi-stable hinge and hinge moments due to aerodynamic loads

As shown in Fig. 7, the lift slope also decreased due to the decambering caused by the initial deflection of the hinge. Post snap-through for hinge deflections of $\delta = -6^\circ$ to $\delta = -7^\circ$, the bi-stable hinge case aligns very closely with the rigid $\delta = -7^\circ$ case. This shows promise for allowing the prediction of post-snap-through C_l from rigid deflection data. The

lift coefficient reduced by $\Delta C_l = 0.178$ immediately post-snap-through, compared to the baseline $\delta = 0^\circ$ case.

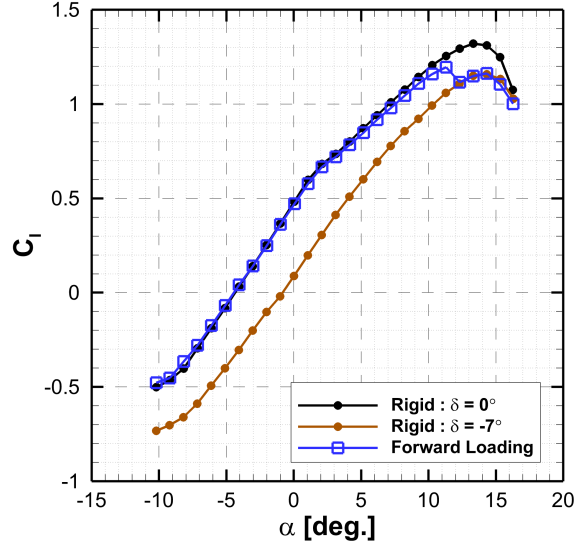
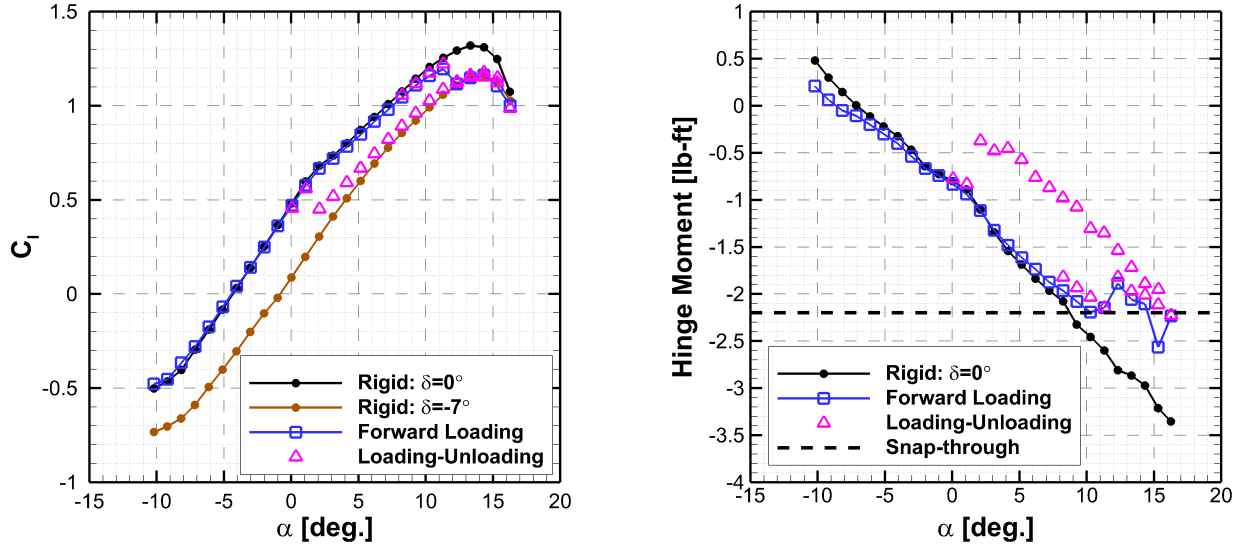


Fig. 7 Lift coefficient of bi-stable hinge test

The reverse snap-through conditions were also investigated. Figure 8 shows the comparison of the forward and reverse snap-through and rigid deflection cases. Forward snap-through occurs at the design point torque of $H' = -2.2$ lb-ft. For the unloading case, α is decreased after $\alpha = 16^\circ$ until reverse snap-through occurs. Shown in Fig. 8a, the lift curve corresponding to the bi-stable hinge demonstrates an increase in airfoil camber as the angle of attack is reduced during the unloading process. This indicates that the hinge is experiencing some initial deflection during the unloading until the reverse snap-through torque is reached.



(a) Forward and reverse snap-through comparison with rigid deflections

(b) Forward and reverse snap-through hinge moment comparison

Fig. 8 Loading and unloading of the bi-stable hinge to show forward and reverse snap-through

One of the goals of this study was to predict when snap-through would occur. This would be valuable for future

applications as it would show that the hinge would deflect at the design point α and C_h . However, the two way coupling of the bi-stable hinge material and the aerodynamic performance coefficients complicate this prediction. Initial work has been done to predict snap-through using hinge moment as a function of deflection angle for the hinge-only tests, shown in Fig. 2b, and hinge moment as a function of deflection for the rigid deflection tests. This was done by selecting an initial deflection, linearly interpolating the rigid deflection results to find the output hinge moment, which in turn was used to find the deflection of the hinge from the hinge-only test results. This was done iteratively until the deflection angle and hinge moment converged. The results are shown in Fig. 9. In Fig. 9b, the linear interpolation shows strong agreement of the predicted deflection angle with the experimental bi-stable hinge before snap-through. However, the linear interpolation predicts snap-through occurring at a lower α than observed experimentally. This indicates that additional closure to the two-way coupling is necessary.

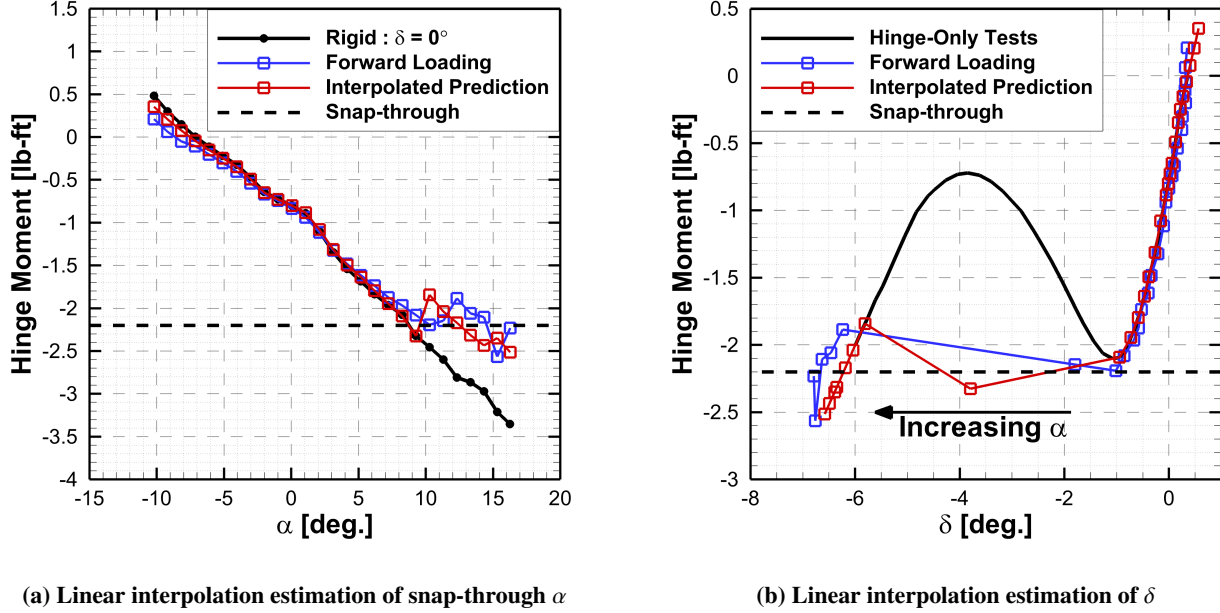


Fig. 9 Linear interpolation estimation of hinge moment and deflection from rigid and hinge-only data

VI. Conclusions and Future Work

An airfoil decambering design using a bi-stable hinge for passive flap deflection was tested for gust load alleviation applications. It was found that the hinge snap-through successfully occurred within the expected limits, showing that the energy wells of the bi-stable hinge can be utilized for airfoil decambering. From these initial results, further work will be done to improve the understanding of the coupling between the bi-stable hinge and the aerodynamic performance coefficients such that snap-through conditions can be predicted. Additionally, the influence of multiple bi-stable hinges in order to achieve multiple deflections will be investigated within the full manuscript.

Appendix

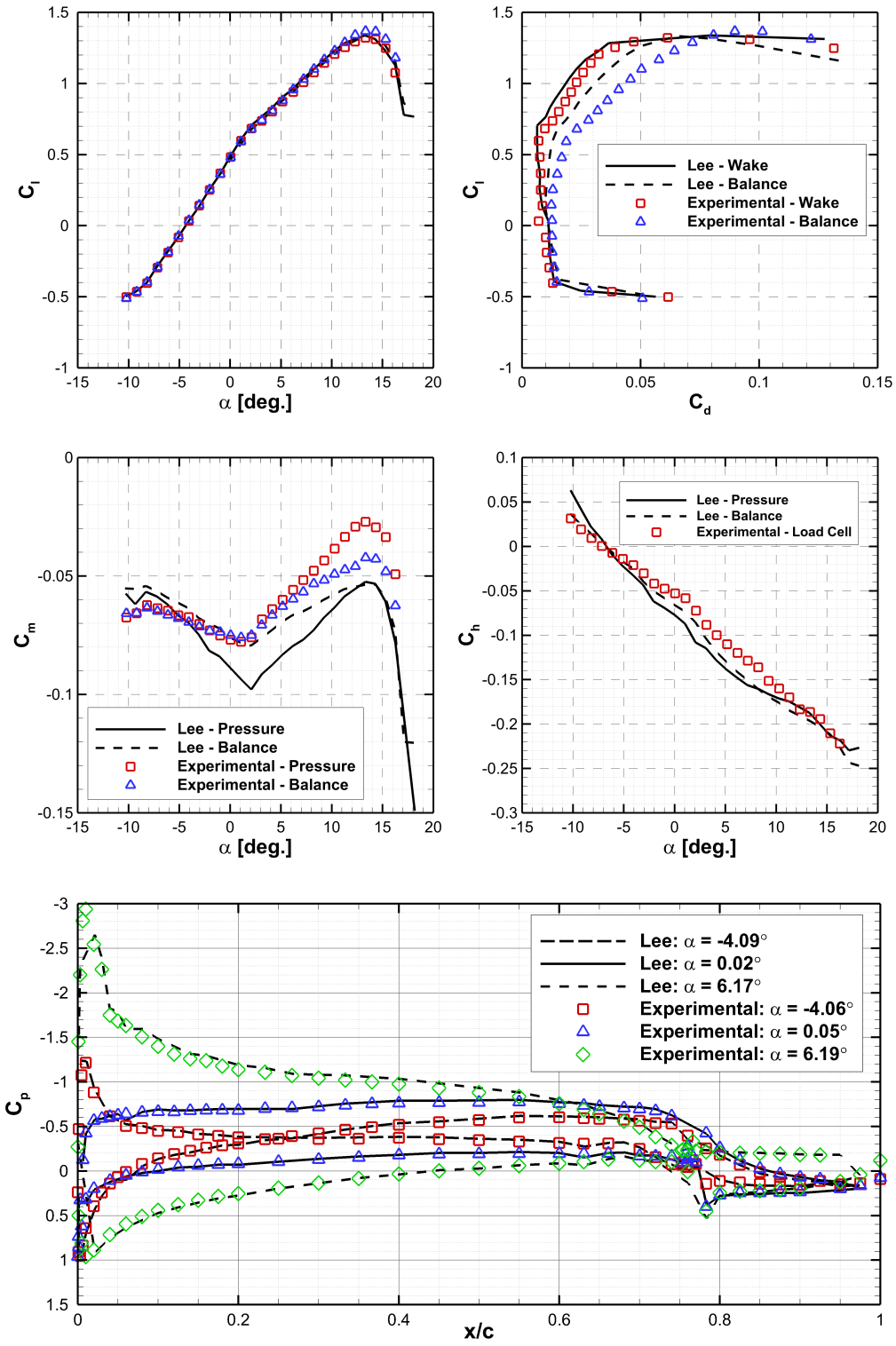


Fig. 10 Surface pressure and force balance data compared with past results from Lee [19]

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